

C5 - Thermal stresses in laminates

Engineering teaching model | 7 October 2026 | SI internally

Uniform temperature change produces thermal resultants NT and MT. ABD solves the free-plate response; local stresses use total strain minus the ply thermal strain. The first-ply factor holds thermal stress fixed and scales the mechanical loads using Maximum Stress and Tsai-Wu.

Graphite/Epoxy (T300/5208)

$\alpha_1 = 0.02$, $\alpha_2 = 22.5$ micrometres/(metre K); $\alpha_{12} = 0$ in principal axes. Reference = 121.111 deg C. Measured stress-free reference selected for residual-stress analysis (250 °F); nominal cure was 350 °F.

The example uses [0/90]_s, illustrative ply thickness 0.125 mm, and $N_x = 100$ kN/m. Final temperatures 20 deg C and -196 deg C are chosen scenario inputs, not measured validation data.

Final T [C]	dT [K]	0-ply residual sigma2 [MPa]	Mechanical factor
20	-101.111	21.503	0.854
-196	-317.111	67.438	0.000

A factor of zero means cooling alone reaches or exceeds the first-ply screening limit. This is reported as calculated, without adjusting properties to obtain agreement.

CTEs: C. B. York (2015), Influence of Bending-Twisting Coupling on Compression Buckling Strength, Table 2. Source document. Temperature: Kriz, Stinchcomb & Tenney, NASA-CR-162921 (1980): measured range 250–300 °F; 250 °F selected for analysis. Source document.

Glass/Epoxy (Scotchply 1002)

$\alpha_1 = 8.6$, $\alpha_2 = 22.1$ micrometres/(metre K); $\alpha_{12} = 0$ in principal axes. Reference = 165.556 deg C. Manufacturer press-cure temperature (330 °F); no measured stress-free temperature was found.

The example uses [0/90]_s, illustrative ply thickness 0.125 mm, and $N_x = 100$ kN/m. Final temperatures 20 deg C and -196 deg C are chosen scenario inputs, not measured validation data.

Final T [C]	dT [K]	0-ply residual sigma2 [MPa]	Mechanical factor
20	-145.556	12.258	0.262
-196	-361.556	30.450	0.000

A factor of zero means cooling alone reaches or exceeds the first-ply screening limit. This is reported as calculated, without adjusting properties to obtain agreement.

CTEs: C. B. York (2015), Influence of Bending-Twisting Coupling on Compression Buckling Strength, Table 2. Source document. Temperature: 3M, Scotchply Reinforced Plastic Type 1002 technical data, §1.8 cure cycles: press cure at 330 °F. Source document.

Limits and verification

Moisture, creep, and temperature-dependent properties are not modelled. Cure chemistry, chemical shrinkage and interlaminar stress are excluded. Reference temperatures depend on cure history; the Scotchply cure temperature is used as an assumed stress-free reference. Cryogenic results extrapolate constant room-temperature properties and are not validated cryogenic strength predictions. Acceptance tests cover ply CTE recovery, symmetry, equal quasi-isotropic expansion, transverse residual tension after cooling, net force balance and mechanical-load scaling.